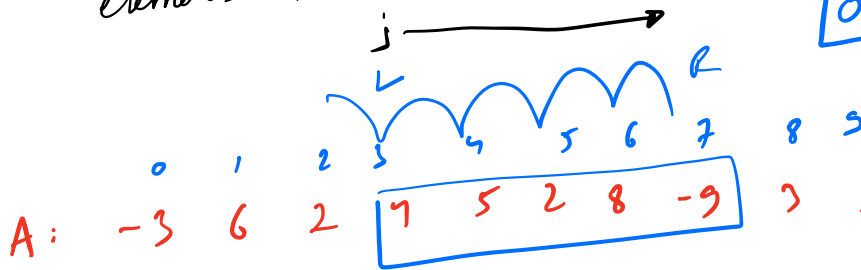


① Prefix Sums

Given an array of N elements & Q queries. [L, R]
 For each query, calculate the sum of all elements in the index range $[L, R]$
 $0 \leq L \leq R < N$



$q = 4$

L	R	Sum
4	8	9
6	9	3
0	4	14
7	7	-9

```

f(i=0; i < Q; i++) {
    // [L, R]
    sum = 0;
    f(j=L; j <= R; j++) {
        sum += A[j];
    }
    // sum is the sum [L, R]
}
    
```

Q (points to $i < Q$)
 N (points to $j <= R$)

$\times$ INT $\rightarrow \sim 10^9$
 $\checkmark$ LONG $\rightarrow \sim 10^{18}$

10^{15}

$1 \leq A[i] \leq 10^9$

TC = $O(QN)$

SC = $O(1)$

$1 \leq N \leq 10^6$
 $1 \leq Q \leq 10^5$

#ops $\rightarrow 10^6 \cdot 10^5 = 10^{11}$

TLE! ~ 1000 sec

- Q. Given the score of the first 10 overs of batting!
After every over, current score $\rightarrow$ given!

Overs:	1	2	3	4	5	6	7	8	9	10
Score:	2	8	14	22	31	49	65	79	89	97

- Q. Total runs scored in the 7th over? $[7, 7]$

$$s[7] - s[6] = 65 - 49 = 16$$

- Q. Last 5 overs? $[6, 10]$

$$s[10] - s[5] = 97 - 31 = 66$$

- Q. Run: 3 5 2 0 5 3
Score: 3 8 10 10 15 18

$$s[6] - s[2] = 18 - 8 = 10$$

$$\text{runs}[L, R] = s[R] - s[L-1]$$

① Cumulative sum for start → Prefix Sum

	0	1	2	3	4
A:	2	5	7	3	2
PS:	2	7	14	17	19

$$PS[3] \rightarrow \text{sum}[0, 3]$$

$$PS[i] \rightarrow \text{sum}[0, i]$$

	0	1	2	3	4	5	6	7	8	9
A:	-3	6	2	4	5	2	8	-9	3	1
PS:	-3	3	5	9	14	16	24	15	18	19

$$\text{sum}[4, 8]: \quad \text{sum}[0, 8] = \text{sum}[0, 3] + \text{sum}[4, 8]$$

$$\text{sum}[4, 8] = \text{sum}[0, 8] - \text{sum}[0, 3]$$

$$\text{sum}[L, R] = \text{sum}[0, R] - \text{sum}[0, L-1]$$

$$\text{sum}[L, R] = PS[R] - PS[L-1]$$

eg: $\text{sum}[4, 8] = PS[8] - PS[4-1]$

$$18 - 9 = 9$$

$$L == 0$$

$$\text{sum}[0, 3] = \text{ps}[3] - \text{ps}[-1]$$

$$\text{sum}(0, i) \rightarrow \text{ps}[i]$$

Q Given an Array A. Build the PS array!

$$\text{ps}[0] = A[0]$$

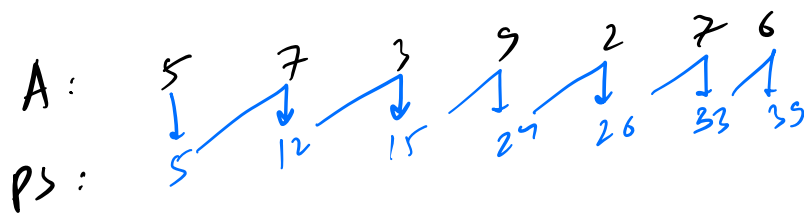
$$\text{ps}[1] = A[0] + A[1]$$

$$\text{ps}[2] = A[0] + A[1] + A[2]$$

$$\text{ps}[3] = A[0] + A[1] + A[2] + A[3]$$

$$\rightarrow \text{ps}[0] \rightarrow A[0]$$

$$i > 0 \quad \boxed{\text{ps}[i] = \text{ps}[i-1] + A[i]}$$



✓ 1. Build the PS []

2. Answer queries

// A[], N, Q.

int ps[N];

ps[0] = A[0]

f(i = 1 → N-1) { $\rightarrow N \times O(1)$
 $ps[i] = ps[i-1] + A[i]$ +

}
 f(i = 0; i < Q; i++) { $\rightarrow Q \times O(1)$
 // [L, R]
 if (L == 0)
 sum = ps[R]
 else
 sum = ps[R] - ps[L-1];
 // sum → Ans of query

$N \leq 10^5$

$Q \leq 10^6$

TC = $O(N + Q)$

$10^5 + 10^6 = \sim 10^6$

SC = $O(N)$

→ ps[]

A: 1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20

ps: 1 4 9 16 25 36 49 64 81 100 121 144 169 196 225 256 289 324 361 400

f(i = 1 → N-1)
 $A[i] += A[i-1];$

ps[]

① Immutable A[]

SC: $O(N)$

② Restore it back!

✓

SC: $O(1)$

WAY 1

TC: $O(N^2)$

op $\rightarrow 10^{11}$

1000 sec

WAY 2

TC: $O(N+Q)$

$\downarrow$
 $\sim 10^6$ ops

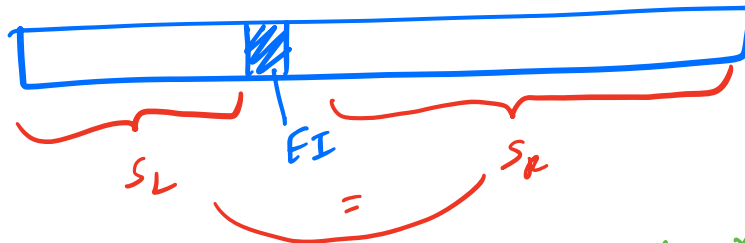
< 1 sec

Q

Equilibrium Index $\rightarrow$ [EI]

Given an array of N elements.
Count the no. of EI.

EI $\rightarrow$ if sum of all elements before it
= after -



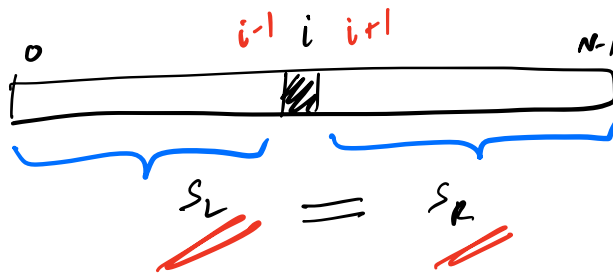
A:

x	x	x	x	x	x	x	x	
0	1	2	3	4	5	7	8	$\rightarrow 0$
-7	1	5	4	2	-4	3	0	

A:

x	✓	x	x	✓	x	
0	1	2	3	4	5	
3	0	-2	2	0	3	$\rightarrow 2$

I) BF



cnt = 0;

for (i = 0; i < N; i++) { → N

 s_L = 0;

 for (j = 0; j < i; j++) { → i

 s_L += A[j];

 }

 s_R = 0;

 for (j = i+1; j < N; j++) { → N-i

 s_R += A[j];

 }

 if (s_L == s_R) {

 cnt++;

 }

}
return cnt;

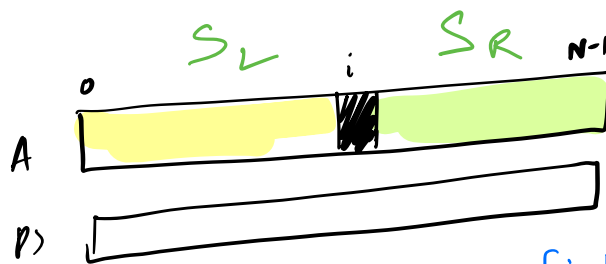
$1 \leq N \leq 10^6$

$TC = O(N \times N)$

$SC = O(1)$

10^{12} ops
↓
 10^9 ms

II)



$$S_L = \text{sum}[0, i-1] = \text{ps}[i-1]$$

$$S_R = \text{sum}[i+1, N-1] = \text{ps}[N-1] - \text{ps}[i]$$

Build the ps[] $\rightarrow O(N)$

```

cnt = 0;
for (i = 0; i < N; i++) {  $\rightarrow N$ 
     $S_L = \text{ps}[i-1];$   $\rightarrow$   $\text{if}(i == 0) S_L = 0$ 
     $S_R = \text{ps}[N-1] - \text{ps}[i];$ 
     $\text{if}(S_L == S_R) \{$ 
        cnt++;
    }
}
cout << cnt;

```

$\rightarrow O(1)$

$2N$

$TC = O(N)$

$SC = O(N)$

$O(1)$

Given an array of size N .
 Given q queries $\rightarrow [L, R]$
 Find the sum of all **EVEN** elements in the index range **$[L, R]$** .

0 1 2 3 4 5 6 7 8 9
 A: 2 3 2 4 2 9 8 7 7 6
 PS: 2 2 4 8 10 18 18 18 24

L	R	Sum
0	5	10
4	4	2
5	5	0

Idea: ASSUME that ODD elements $\rightarrow 0$

```

ps[];
ps[0] = (A[0] % 2 == 0) ? A[0] : 0;
for (i = 1; i < N; i++) {
    ps[i] = ps[i-1] + (A[i] % 2 == 0) ? A[i] : 0;
}
  
```

N

+

q

// Answer given before

$$TC = O(N+g)$$

$$SC = O(N)$$

$O(1)$

Q

Given N array elements & Q queries.

For each query $[L, R] \rightarrow$ find the no. of even elements in $[L, R]$

0 1 2 3 4 5 6 7 8 9
A: [2, 4, 3, 7, 9, 8, 6, 5, 7, 5]

[2, 7] $\rightarrow$ 2

[0, 2] $\rightarrow$ 2

[7, 9] $\rightarrow$ 0

0 1 2 3 4 5 6 7 8 9
A: [2, 4, 3, 7, 9, 8, 6, 5, 7, 5]

EVEN $\rightarrow$ 1
ODD $\rightarrow$ 0

1 1 0 0 0 1 1 0 1 0
P: 1 2 2 2 2 3 4 4 5 5

ANS Q: 5

$O(Q)$

Build

$O(N)$

TC = $O(N + Q)$

SC = $O(N)$

$O(1)$

~~Q~~ Given a string S $S_i \in \{a-z\}$

Given Q queries $\rightarrow [L, R, ch]$

find the # of occurrences of ch in $[L, R]$

$S =$ $\begin{matrix} 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ a & b & c & b & a & d & g \end{matrix}$

0	4	a	$\rightarrow 2$
2	6	g	$\rightarrow 1$
5	6	h	$\rightarrow 0$

INT $\sim 10^9$
LONG $\sim 10^{18}$